

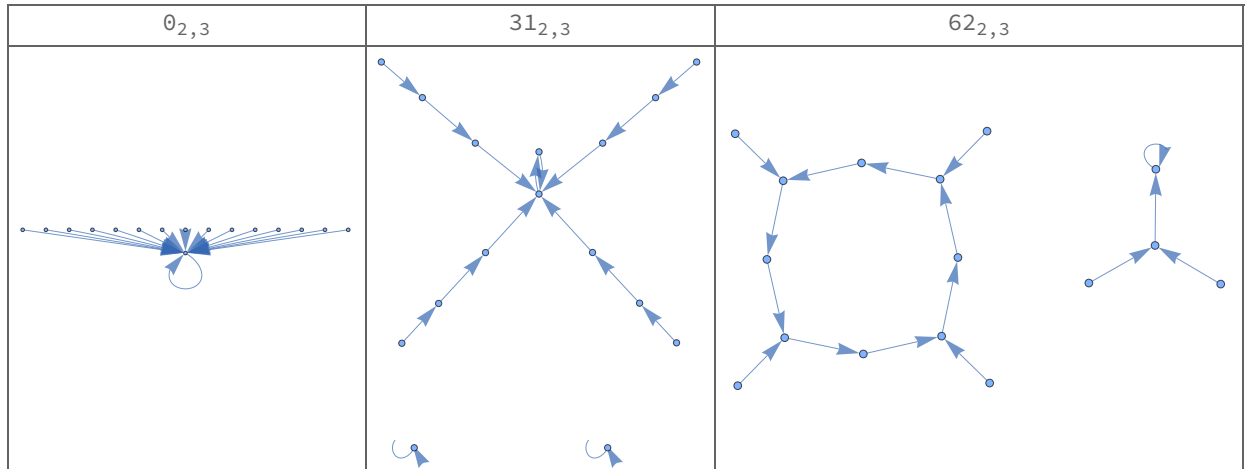
Similarity Index of 1D Cellular Automata

Graph approach

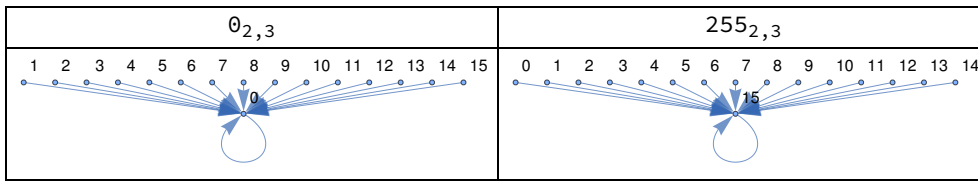
Erick Jesse Angeles Lopez

Graph of space evolution

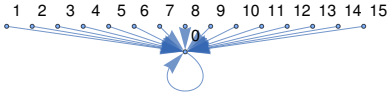
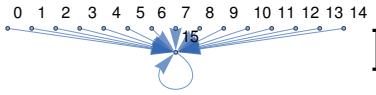
In a space with 4 cells, we have $2^4 = 16$ possible spaces. Each of one can be represented as a node and be connected by a rule:



But we can compare both graphs, because they can have the same shape but not the same nodes:



We can test if they are isomorphic, that means, there is a bijective function that maps each vertex in the transition matrix

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In[ ]:= IsomorphicGraphQ[, ]
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Out[ ]:= True
```

We can obtain the canonical form of this graphs. There is not an specific algorithm:

1. Calculate the degree of each node and en-queue the leaf nodes.

2. For each leaf node, assign a structural code for the parent.

2.1. en-queue the parent node if now it is a leaf.

3. Find cicles and asssing codes

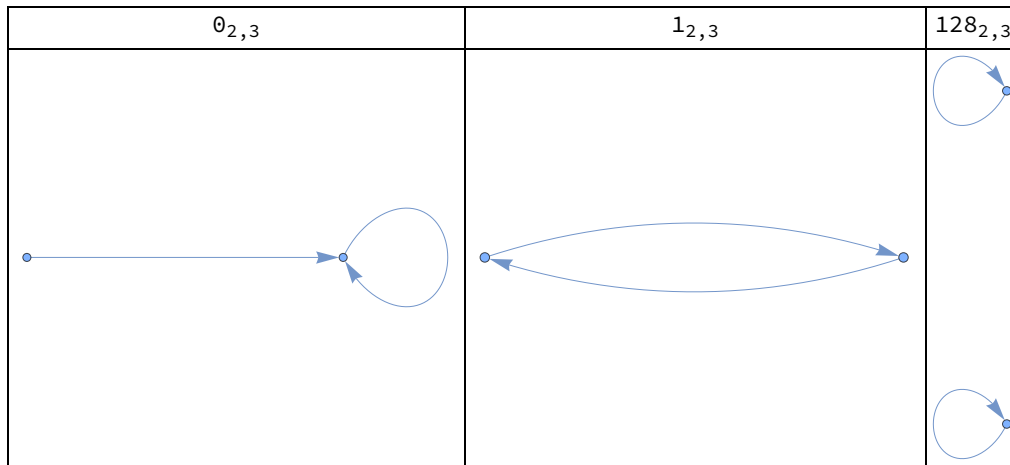
Differences with grouping by mirror and state permutatation:

Out[=]=

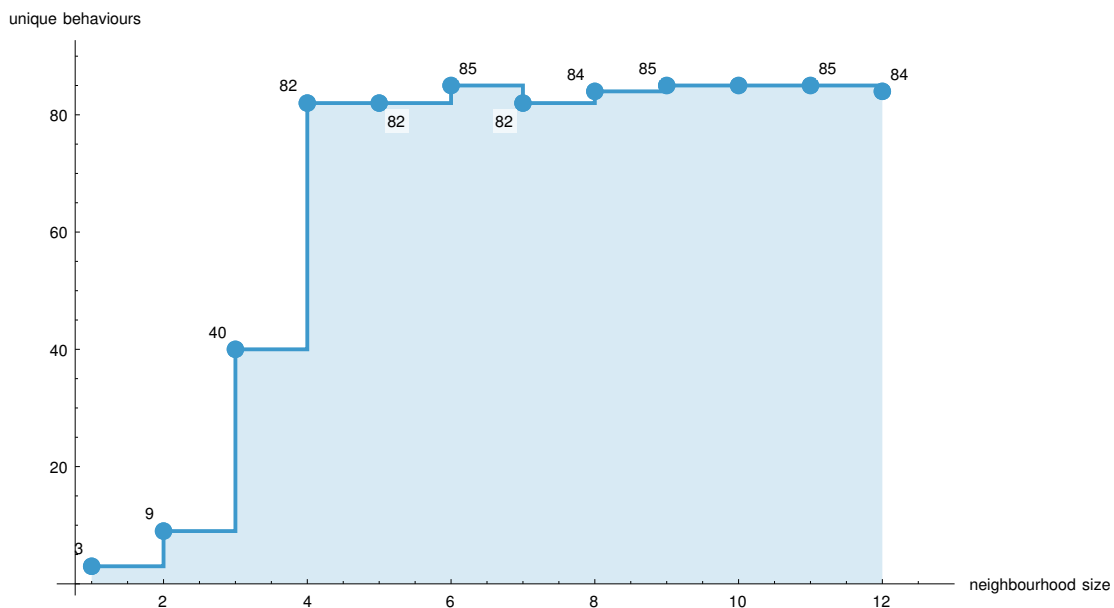
$\{15_{2,3}, 85_{2,3}, 170_{2,3}, 240_{2,3}\}$	$\{\{15_{2,3}, 85_{2,3}\}, \{240_{2,3}, 170_{2,3}\}\}$
$\{23_{2,3}, 178_{2,3}\}$	$\{\{23_{2,3}\}, \{178_{2,3}\}\}$
$\{25_{2,3}, 45_{2,3}, 61_{2,3}, 67_{2,3},$ $75_{2,3}, 89_{2,3}, 101_{2,3}, 103_{2,3}\}$	$\{\{45_{2,3}, 75_{2,3}, 101_{2,3}, 89_{2,3}\},$ $\{67_{2,3}, 61_{2,3}, 25_{2,3}, 103_{2,3}\}\}$
$\{28_{2,3}, 70_{2,3}, 78_{2,3}, 92_{2,3},$ $141_{2,3}, 157_{2,3}, 197_{2,3}, 199_{2,3}\}$	$\{\{28_{2,3}, 199_{2,3}, 70_{2,3}, 157_{2,3}\},$ $\{78_{2,3}, 141_{2,3}, 92_{2,3}, 197_{2,3}\}\}$
$\{77_{2,3}, 232_{2,3}\}$	$\{\{77_{2,3}\}, \{232_{2,3}\}\}$
$\{105_{2,3}, 150_{2,3}\}$	$\{\{105_{2,3}\}, \{150_{2,3}\}\}$

The mirror and state permutation have the same isomorphic graph. But here comes the next question: What should be the size of the space?

Because if we considered a single cell, we have only 3 behaviors:



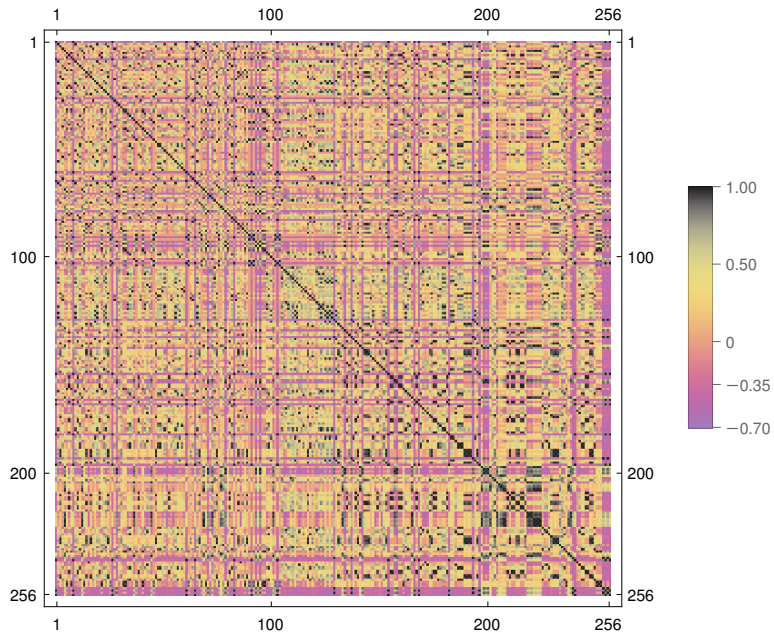
Relation between the space size and the unique behaviors:



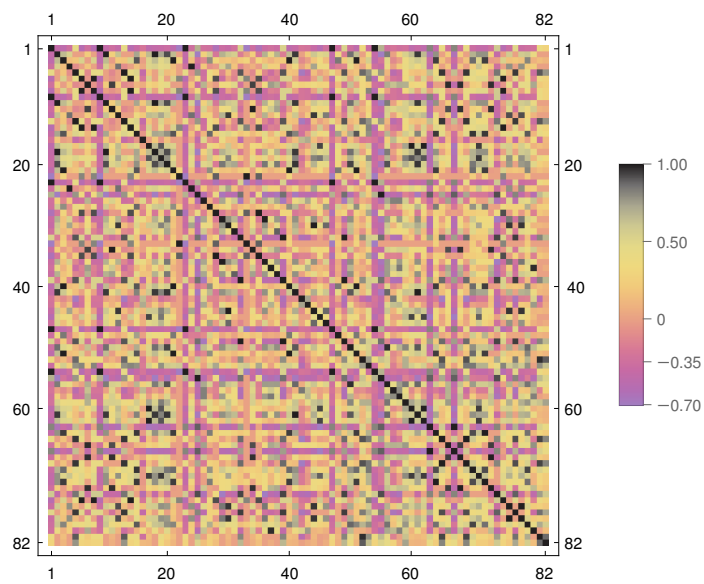
If we increase the space size we find similarities between 24 rules, but none of them are consistent along the space size.

Meassure the similarity

To measure the similarity between rules. We obtain the cosine distances between the eigen values of each transition matrix:



Similarity matrix for the unique behaviors:



All distances between the mirror and inverse rule sets are zero; that is, they are equivalent with respect to the eigenvalues of the transition matrix in a state space of size 4.

Thanks